REAL-TIME DEVOPS SRE

OBSERVABILITY & RELIABILITY PRODUCTION PROJECT

Hands-on flow

Docker Compose → OpenTelemetry → Prometheus/Loki/Tempo → Grafana/Alertmanager → k6 → incidents → Kubernetes/AKS → GitOps/Argo CD

Golden Signals

Latency, Traffic, Errors and Saturation.

Practice SLOs

API availability 99.9%; API latency under 500 ms for 99%; order success 99.9%; payment success 99.5%.

Incident workflow

Alert → Metrics → Logs → Traces → Root Cause → Mitigation → Validation → Postmortem.

First hands-on session

```
cp .env.example .env

docker compose build

docker compose up -d

curl http://localhost:8000/health

k6 run load-tests/smoke.js

Open Grafana at http://localhost:3000

Inspect Prometheus → Loki → Tempo

Complete incidents/INC-001-order-api-latency.md
```